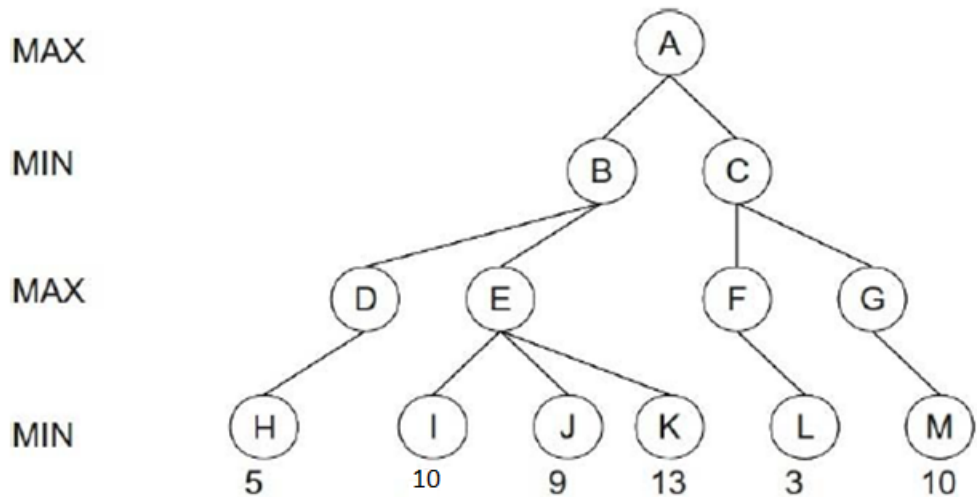


Practice Problem Games

Question 1

Perform **alpha-beta pruning** on the following game tree. Write the nodes, in the order they are assigned minimax value, in the table below. Also write the minimax value assigned to each node.



Your Answers

Minimax

Nodes*	H	D	I	J	K	E	B	L	F	M	G	C	A
Minimax Value	5	5	10	9	13	13	5	3	3	10	10	3	5

*Write nodes in the order they are assigned minimax value

Alpha-Beta pruning

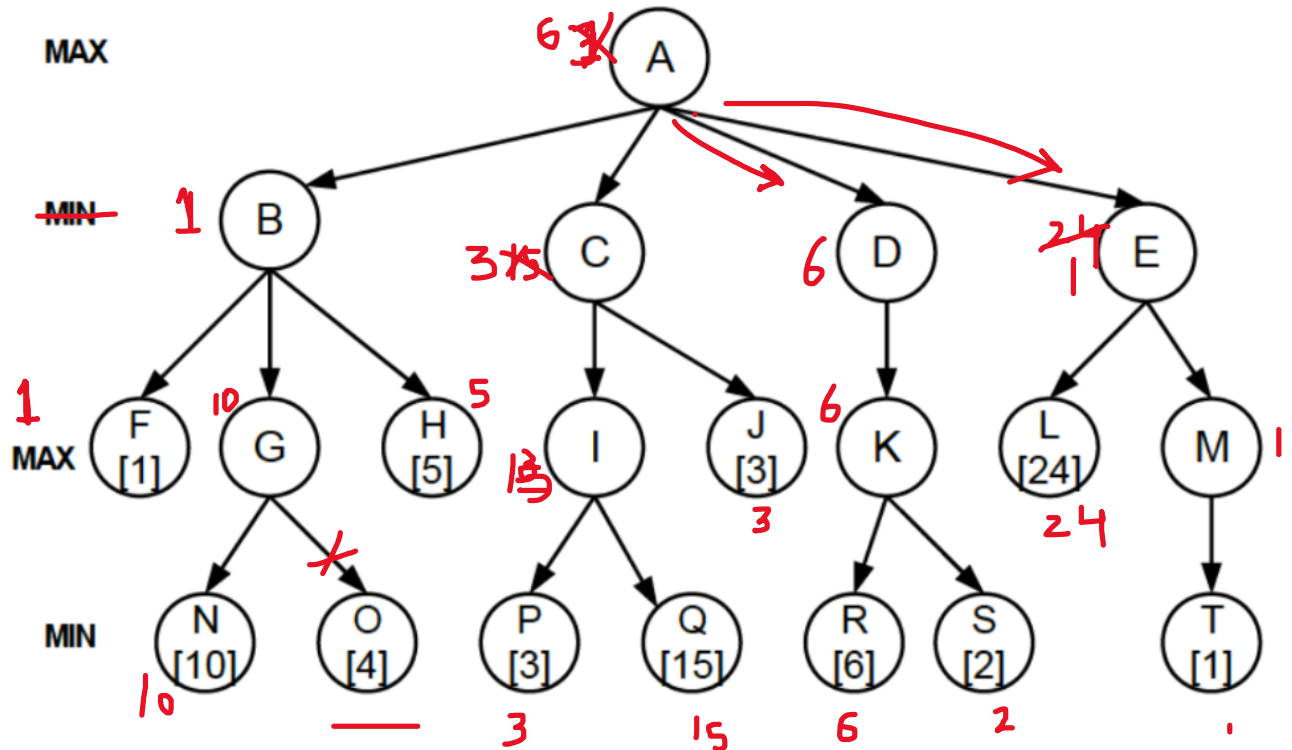
Nodes*	H	D	I	E	B	L	F	C	A				
Minimax Value	5	5	10	10	5	3	3	3	5				

*Write nodes in the order they are assigned minimax value

Question 2

Use minimax and alpha-beta pruning and expectimax algorithms to solve the following problem.

List the nodes along with the minimax value in the order they are assigned minimax value.



Solution:

Minimax:

F	N	O	G	H	B	P	Q	I	J	C	R	S	K	D	L	T	M	E	A		
1	10	4	10	5	1	3	15	15	3	3	6	2	6	6	24	1	1	1	6		

Alpha-beta pruning:

F	N	G	H	B	P	Q	I	J	C	R	S	K	D	L	T	M	E	A		
1	10	>=10	5	1	3	15	15	3	3	6	2	6	6	24	1	1	1	6		

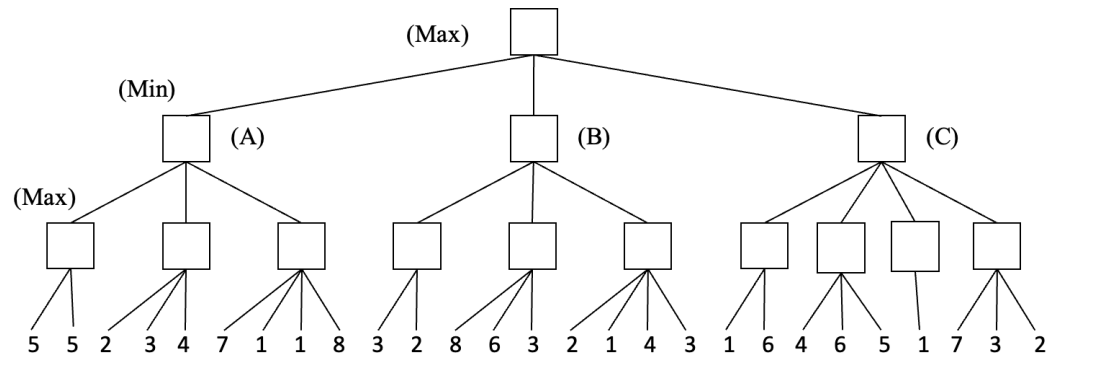
Expectimax:

F	N	O	G	H	B	P	Q	I	J	C	R	S	K	D	L	T	M	E	A		
1	10	4	10	5	5.3	3	15	15	3	9	6	2	6	6	24	1	1	12.5	12.5		

In the game tree below, the values below the nodes are the static evaluations at those nodes. MAX next to a horizontal line of nodes means that the maximizer is choosing on that turn, and MIN means that the minimizer is choosing on that turn.



Question 4. In the game tree below, the value below each leaf node is the utility value at that node. Perform a minimax search with **alpha-eta pruning** and answer the following questions.



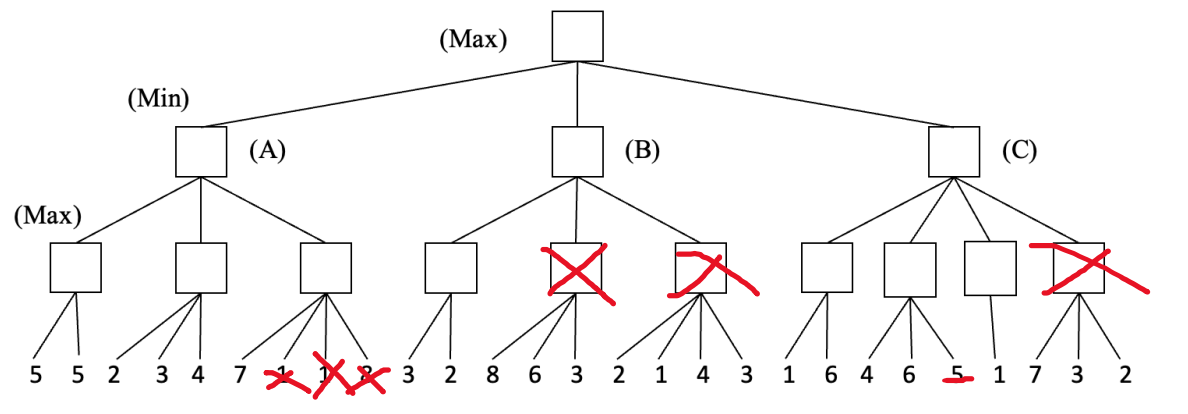
[1 pts] What is the best move for **Max**? (Write A, B or C)

A

[1 pts] What score does **Max** expect to achieve?

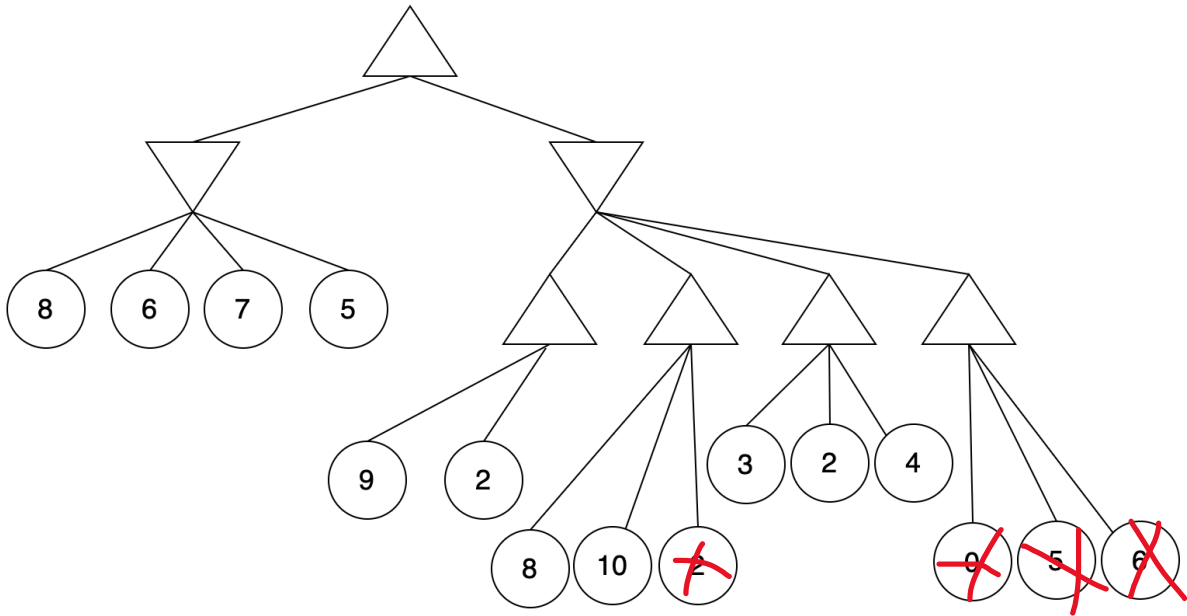
4

[8 pts] Cross out nodes in the figure below that will be pruned by Alpha-Beta Pruning. **Mark each node that will be pruned.**



Question 5.

Consider the minimax game tree shown below. The triangles pointing up, such as the root, correspond to the MAX player, while the triangles pointing down correspond to the MIN player. The value written inside the terminal nodes is the utility at that node.



1. [2 pt] What is the minimax value of the root?

5

2. Draw an X through all of the nodes in the figure above that would be pruned (i.e. not explored at all) by α - β pruning.